Effat University

Evaluating the Generalizability of Heart Disease Prediction Models Across Heterogeneous Populations

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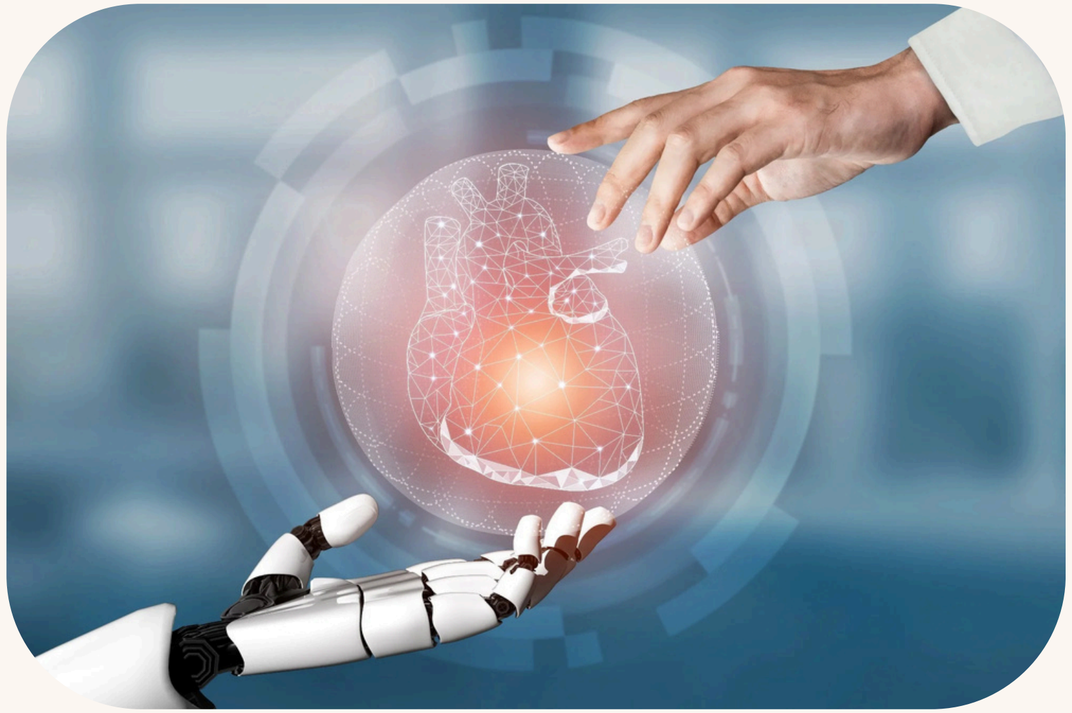
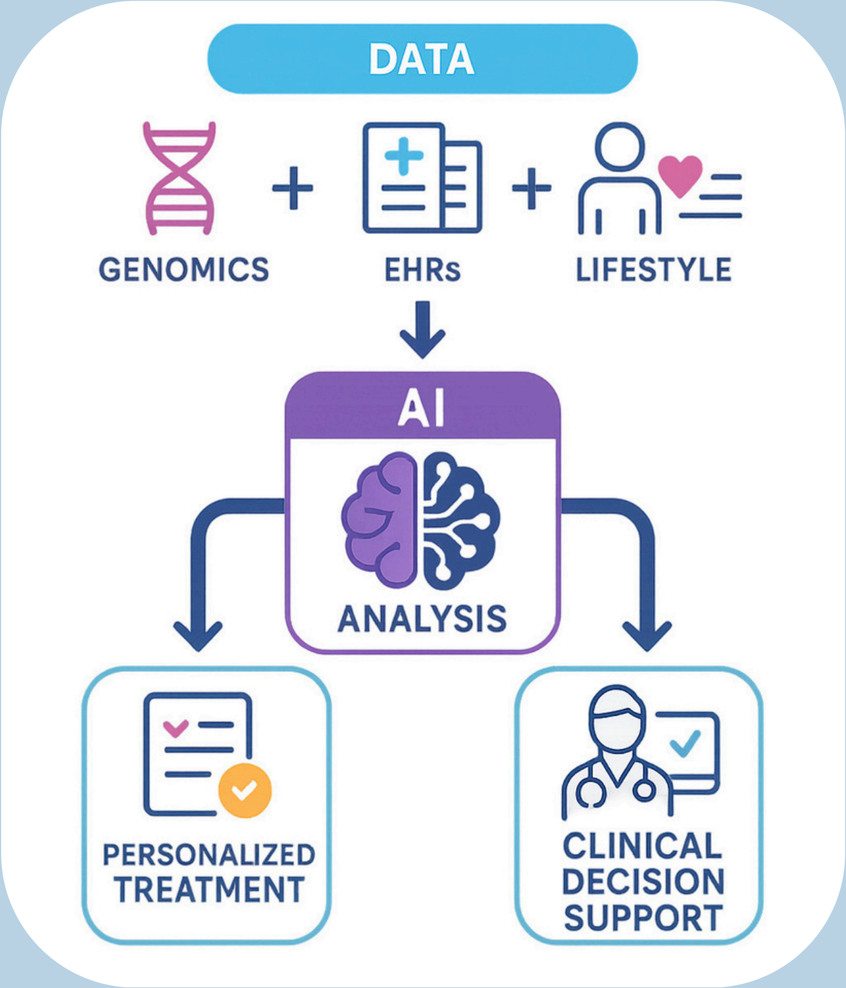
Data Science - Project Presentation



Problem Overview

- Heart disease is a leading cause of death worldwide
- Machine learning models are widely used for prediction
- Many models report high predictive accuracy
- Most evaluations use the same dataset
- Real-world performance remains uncertain

Do these models generalize to new populations?



Sources: BMJ (2023); JAMA Internal Medicine (2021); JAMA Network Open (2025)

Alignment with SDGs & Vision 2030

- **SDG 3 Good Health and Well-being:** Improving heart disease prediction and healthcare outcomes
- **SDG 9 Industry, Innovation and Infrastructure:** Applying machine learning in healthcare systems
- **SDG 10 Reduced Inequalities:** Ensuring reliable model performance across populations

- **Saudi Vision 2030:** Advancing digital health and improving healthcare quality through AI

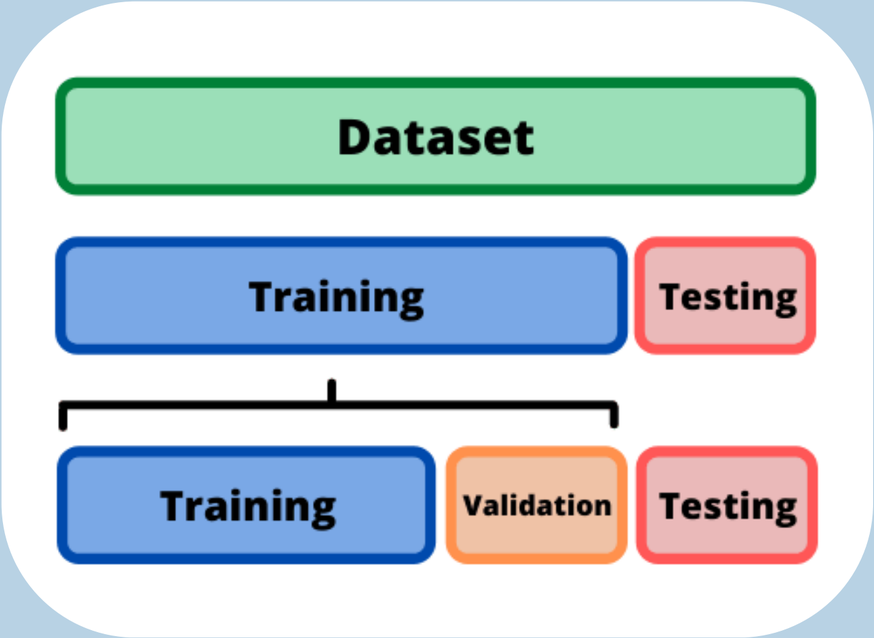
VISION رؤيـــة 2030
المملكة العربية السعودية
KINGDOM OF SAUDI ARABIA

Sources: JAMA Cardiology (2021); JACC (2024); Frontiers in Digital Health (2025)

Research Gap

- Many studies rely on a single dataset
- Most evaluations rely on simple train–test splits (internal validation)
- Limited evaluation across different populations
- Model generalization across cohorts is not well studied

Lack of systematic cross-cohort evaluation



Sources: BMJ (2023); BMC Medical Research Methodology (2025); JAMA Network Open (2025)

Research Questions & Hypotheses

RQ1:

How does performance differ between internal and external validation?

RQ2:

How does model performance change across different cohorts?

RQ3:

Which model generalizes better across populations?

H1:

Models will show higher performance under internal validation than external validation.

H2:

Model performance will decrease on external cohorts.

(Not supported by results)

H3:

Ensemble models will be more robust than simpler models across populations.

Sources: JAMA Internal Medicine (2021); JAMA Network Open (2025); npj Digital Medicine (2021)

Dataset

Dataset: UCI Heart Disease Dataset (clinical data collected from hospitals and cardiology centers)

Cohorts: Cleveland , Hungary , Switzerland , VA Long Beach

Task: Binary classification (heart disease presence)

Total samples across 4 cohorts: ~920 patients (Cleveland: 303, Hungary: 294, Switzerland: 123, VA: 200)

Features: 13 selected clinical features (ex: age, chest pain, cholesterol)

Data types: Numerical + Categorical

Missing values: Present, handled during preprocessing

Enables evaluation of model generalization across diverse patient populations

Source: UCI Machine Learning Repository (Dua & Graff, 2019)

Data Cleaning (Preprocessing).

Step 1) Data cleaning:
- Converted features to number format
- Fixed invalid categorical values
- Removed inconsistent entries

Step 2) Missing values handling:
- Numerical → median imputation
- Categorical → most frequent value

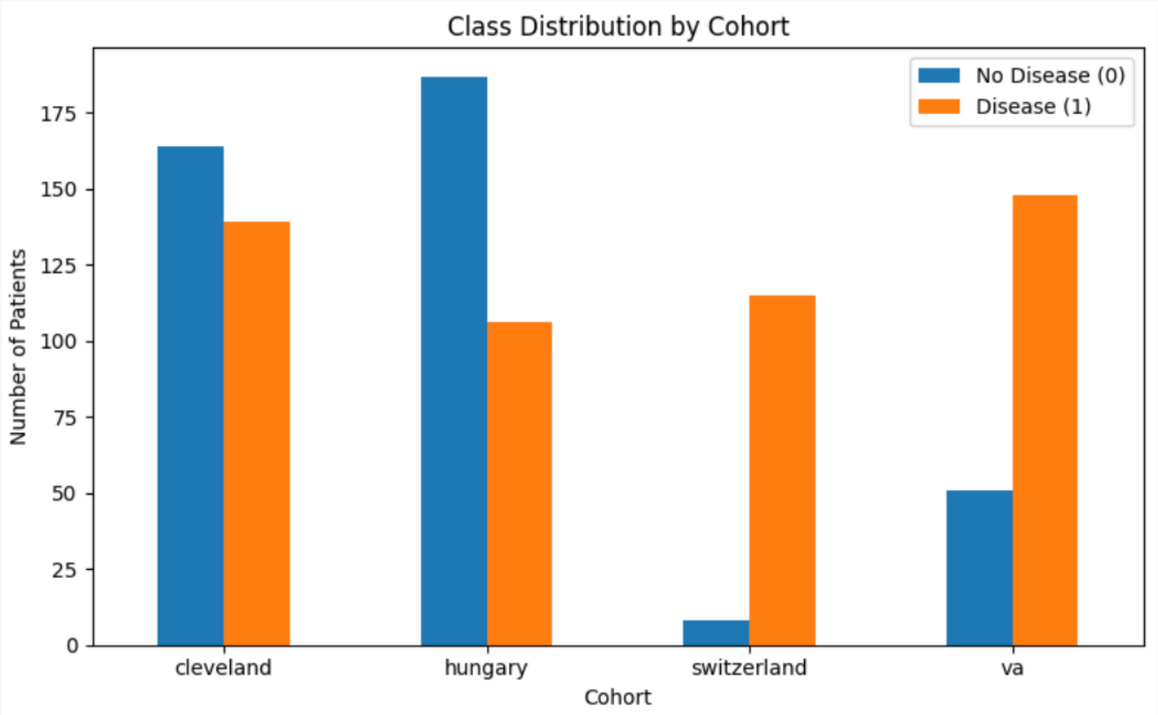
Step 3) Feature selection:
- Kept stable clinical features
- Removed unstable/high-missing features (ex: ca, chol, thal, fbs)

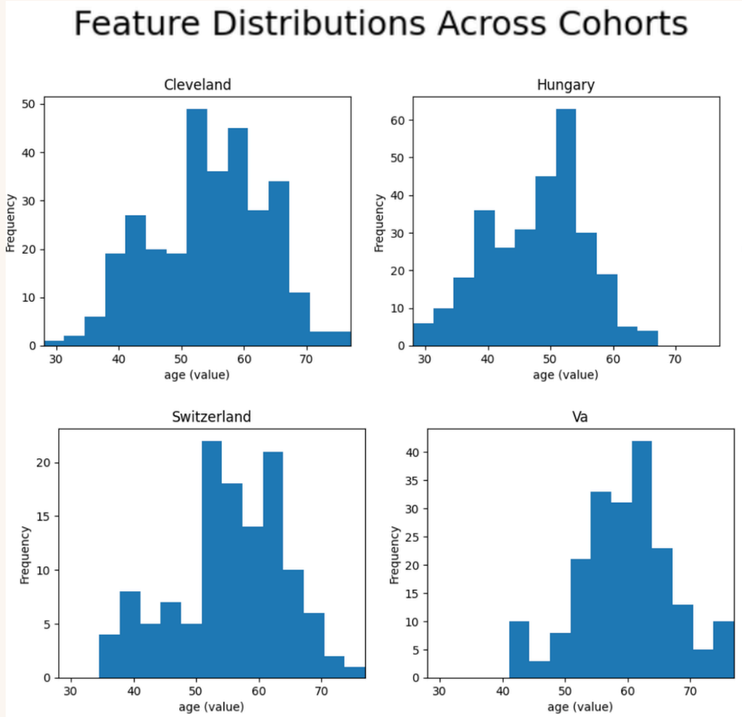
Step 4) Outlier handling:
- Applied IQR-based capping

Step 5) Data transformation:
- Standardized numerical features
- One-hot encoding for categorical variables

Step 6) Pipeline:
- Preprocessing applied within pipeline to prevent data leakage

Exploratory Data Analysis (EDA)





- Analyzed class distribution across cohorts
- Explored key feature distributions (age, thalach)
- Identified missing values and inconsistencies
- Compared patterns across cohorts

Key Findings:

- Class distribution varies across cohorts
- Feature distributions differ between populations
- Key features remain consistent (e.g., thalach, age)
- Variability suggests generalization challenges

Solution

Developed machine learning models for heart disease prediction

Metrics:
Accuracy, Precision, Recall, F1-score, and ROC-AUC

Models Used
- Logistic Regression
- Random Forest

Validation:
- Internal validation (same cohort)
- External validation (different cohorts)

Cross-cohort analysis:
- Training:Cleveland + Hungary
- Testing:Switzerland and VA

Focus on evaluating model generalization across different populations

Sources: JAMA Internal Medicine (2021); JAMA Network Open (2025); npj Digital Medicine (2021); Hastie et al. (2009)

Results

Logistic Regression Results:

External Validation Results

	model	train_cohort	test_cohort	n_train	n_test	accuracy	precision	recall	f1	roc_auc
0	LogisticRegression	cleveland+hungary	switzerland	596	123	0.739837	0.966292	0.747826	0.843137	0.758696
1	LogisticRegression	cleveland+hungary	va	596	199	0.753769	0.861314	0.797297	0.828070	0.757750

Internal Validation Results

	model	cohort	n_total	internal_accuracy	internal_precision	internal_recall	internal_f1	internal_roc_auc
0	LogisticRegression	cleveland	303	0.824176	0.825000	0.785714	0.804878	0.890671
1	LogisticRegression	hungary	293	0.840909	0.800000	0.750000	0.774194	0.925781
2	LogisticRegression	switzerland	123	0.945946	0.945946	1.000000	0.972222	0.557143
3	LogisticRegression	va	199	0.800000	0.800000	0.977778	0.880000	0.765185

Random Forest Results:

External Validation Results

	model	train_cohort	test_cohort	n_train	n_test	accuracy	precision	recall	f1	roc_auc
0	RandomForest	cleveland+hungary	switzerland	596	123	0.642276	0.973333	0.634783	0.768421	0.765217
1	RandomForest	cleveland+hungary	va	596	199	0.648241	0.861111	0.628378	0.726562	0.725755

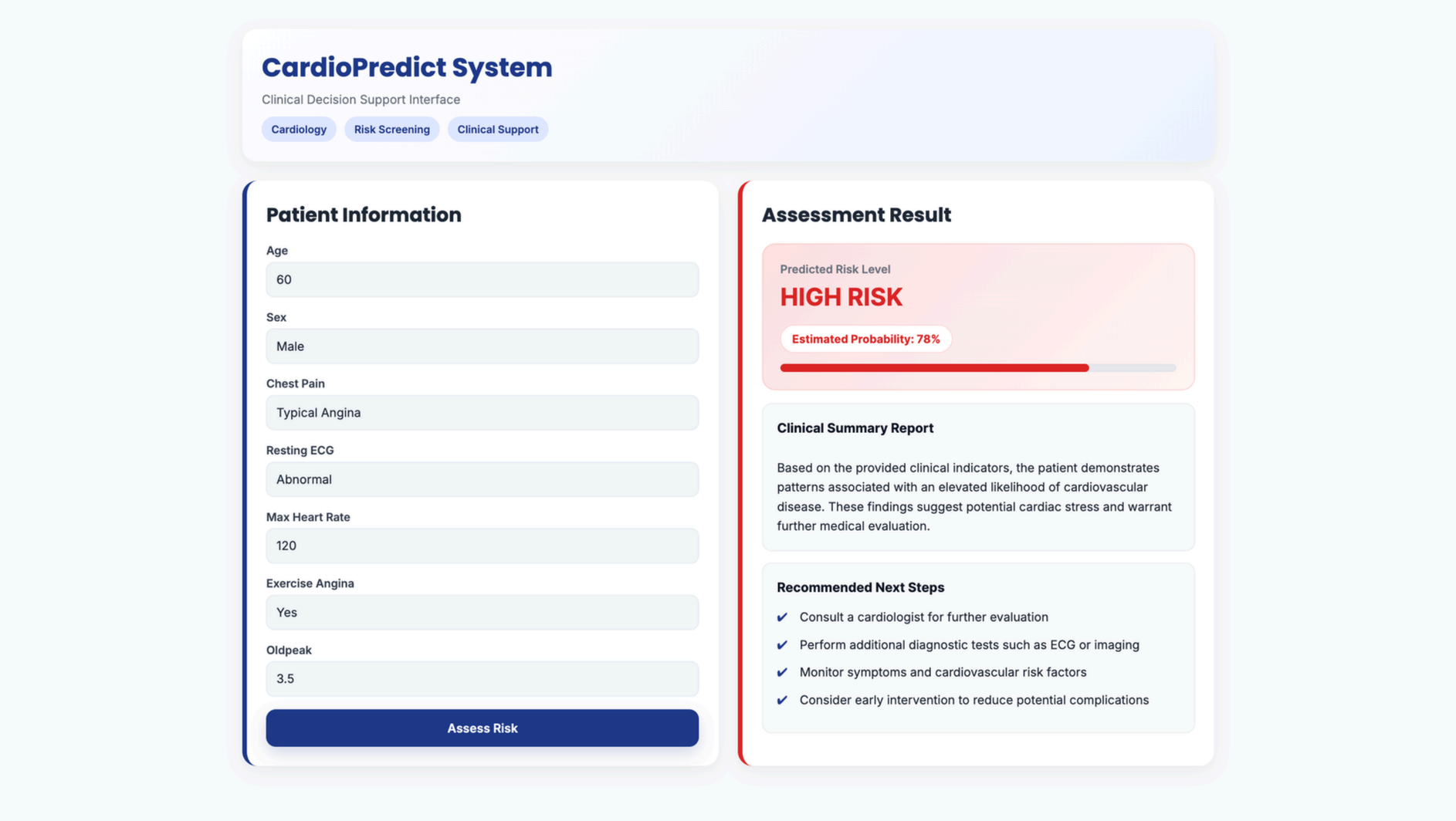
Internal Validation Results

	model	cohort	n_total	internal_accuracy	internal_precision	internal_recall	internal_f1	internal_roc_auc
0	RandomForest	cleveland	303	0.780220	0.761905	0.761905	0.761905	0.878523
1	RandomForest	hungary	293	0.829545	0.742857	0.812500	0.776119	0.915179
2	RandomForest	switzerland	123	0.945946	0.945946	1.000000	0.972222	0.514286
3	RandomForest	va	199	0.783333	0.796296	0.955556	0.868687	0.791852

- Internal validation generally yields higher performance, while external validation shows reduced generalization.
- External validation reveals a noticeable decline across cohorts, indicating distribution shift.
- Logistic Regression demonstrated more stable performance across external cohorts compared to Random Forest.
- Random Forest shows greater sensitivity to cohort variability.
- Performance decreases are likely due to distribution shift between cohorts.
- Some cohorts (ex: Switzerland) show inflated internal performance due to class imbalance.

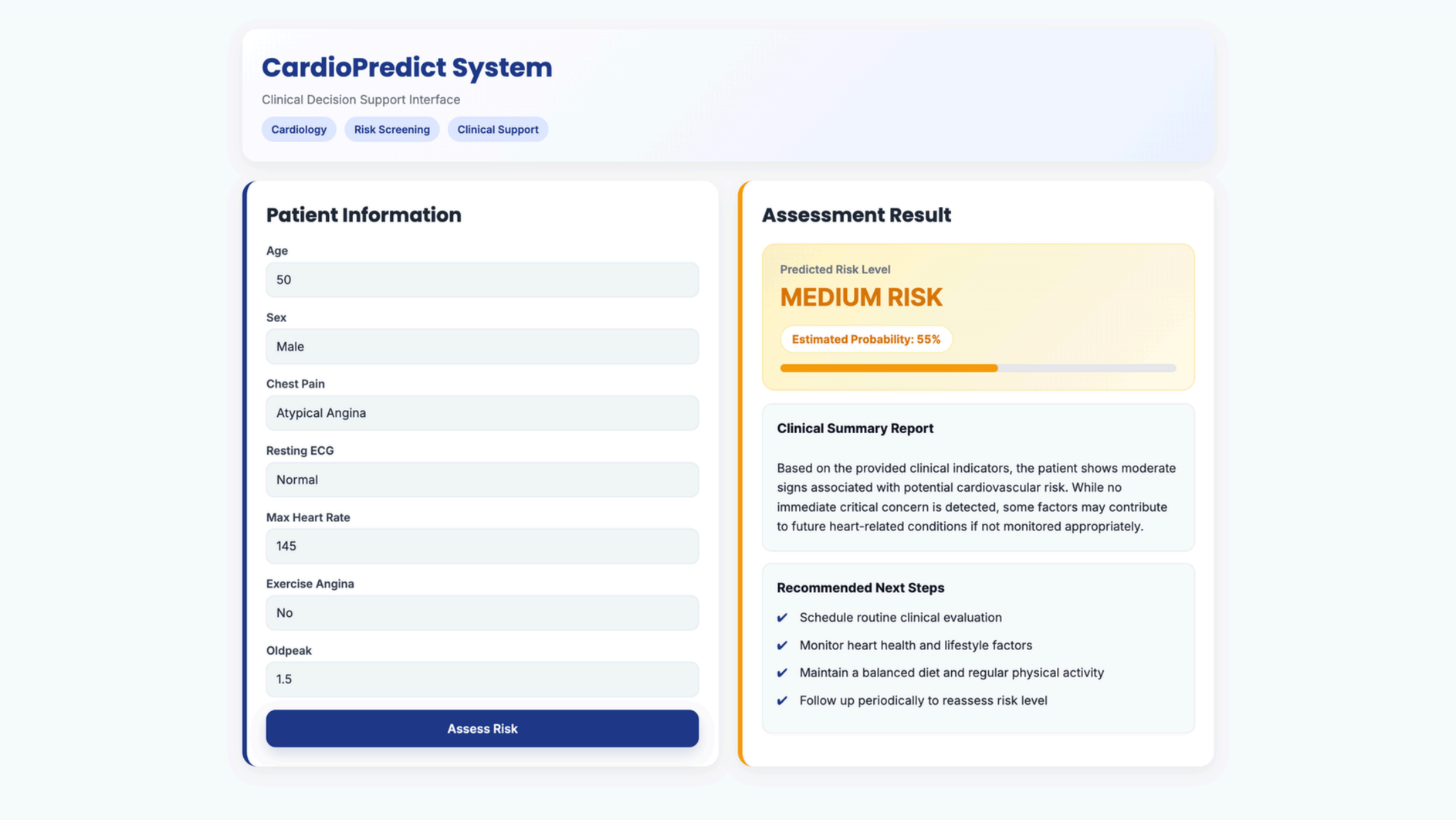
Supports H1 and H2, but contradicts H3

UI/UX Design



A prototype designed to allow clinicians to input patient data and receive a clear, interpretable heart disease risk assessment with actionable recommendations.

UI/UX Design



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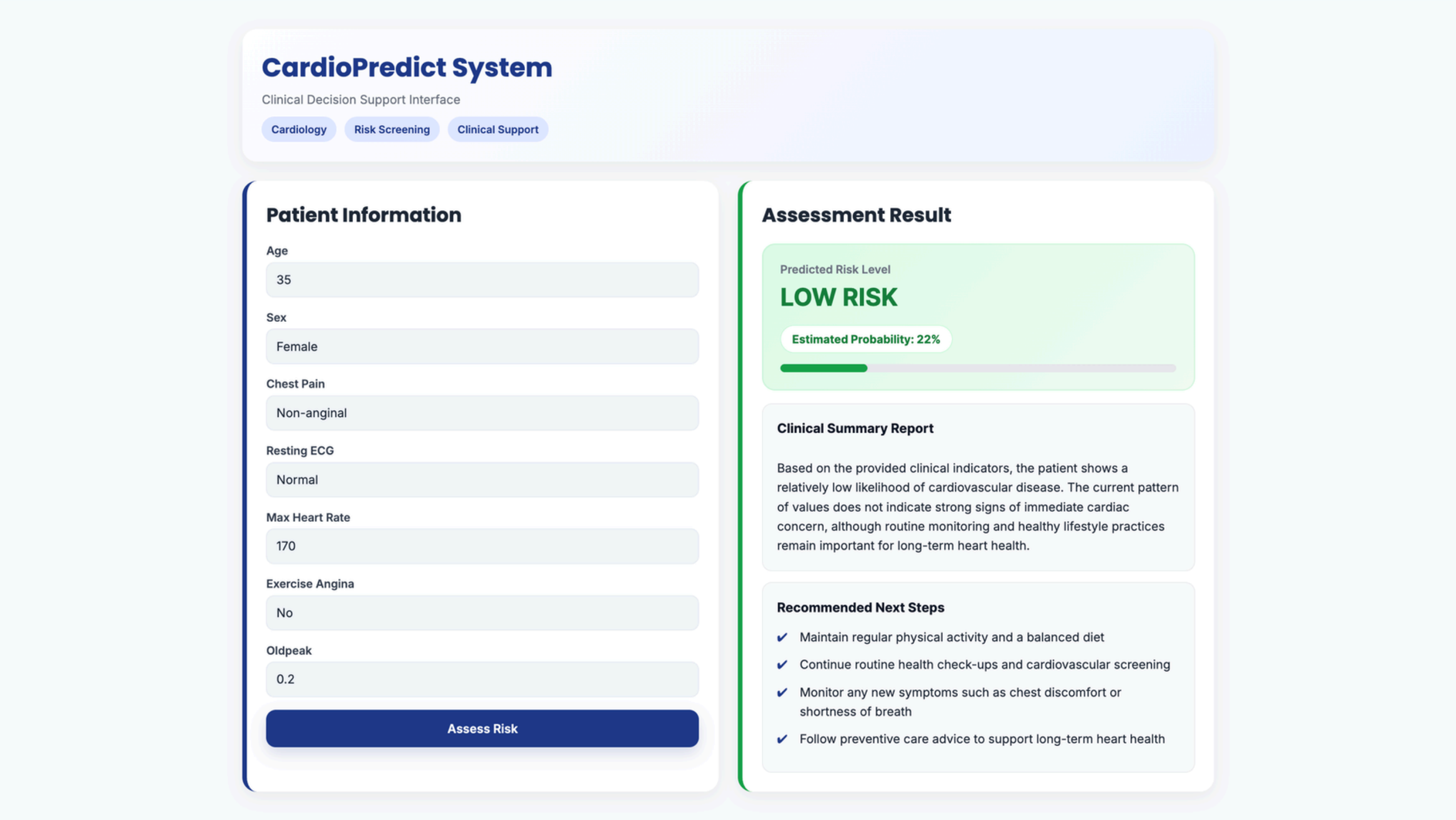

A prototype designed to allow clinicians to input patient data and receive a clear, interpretable heart disease risk assessment with actionable recommendations.

Conclusion

- Machine learning models perform well under internal validation but struggle to generalize across populations
- External validation reveals performance drops due to distribution shift
- Logistic Regression showed more stable performance than Random Forest
- Cross-cohort evaluation is essential for reliable healthcare AI

Sources: BMJ (2023); JAMA Internal Medicine (2021); JAMA Network Open (2025); npj Digital Medicine (2021)

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Thank You For Listening!

Any Questions?